

Dr. Kewei Sha

University of North Texas
College of Information
3940 N Elm St
Denton, TX 76207

Office: Discovery Park E295A
Phone: 940-565-3271
Email: kewei.sha@unt.edu
Web: <https://comersha.github.io>

Research Interests

Applied AI, Edge Computing, Security and Privacy, Smart Healthcare Systems, Blockchain, Internet of Things, and Virtual and Augmented Reality

Employment

Director, Data Science Graduate Program, University of North Texas, 09/2024 –
Director, Data Science Undergraduate Program, University of North Texas, 01/2025-08/2025
Associate Professor of Data Science, University of North Texas, 09/2023 –
Affiliated Professor of Information Science, University of North Texas, 09/2025 –
Professor of Computer Science, University of Houston-Clear Lake, 06/2023 – 08/2023
Associate Professor of Computer Science, University of Houston-Clear Lake, 09/2019 – 06/2023
Associate Director, Cyber Security Institute at UHCL, 09/2015 – 06/2023
Assistant Professor of Computer Science, University of Houston-Clear Lake, 09/2015 – 08/2019
Chair, Dept. of Technology and Software Eng., Oklahoma City University, 07/2013-08/2015
Associate Professor of Computer Science, Oklahoma City University, 07/01/2015 – 08/31/2015
Assistant Professor of Computer Science, Oklahoma City University, 07/2011-06/30/2015
Visiting Assistant Professor of Computer Science, Oklahoma City University, 08/2008-06/2011

Education

Ph.D. in Computer Science, 09/2003-11/2008
Wayne State University, Detroit, MI

M.S. in Computer Science, 09/2003-12/2006
Wayne State University, Detroit, MI

B.S. in Computer Science, 09/1997-07/2001
East China University of Science and Technology, Shanghai, China

Honors and Awards

- Leadership Fellow, University of North Texas, 2025-2026
- Outstanding Service Award from IEEE/ACM CHASE 2025, 2025
- Recognition of Service as General Chair of IEEE MOST 2024, 2024
- Recognition of Service as General Chair of IEEE/ACM SEC 2023, 2023
- University Faculty Fellowship Award, UHCL, 2022
- UHCL Anthony & Barbara Lekkos Endowment Faculty Fellowship, 2021
- Recognition of Service as General Chair of IEEE/ACM CHASE 2021 by ACM, 2021
- Recognition of Service as Program Chair of ACM/IEEE SEC 2020 by ACM, 2020
- UHCL President's Distinguished Research Award, 2019
- IEEE Outstanding Leadership Award from IEEE ICCCN 2019

- Paper “On security challenges and open issues in Internet of Things” is featured in Future Generation Computer Systems’ Research Insights, 2019
- 2018 Albert Nelson Marquis Lifetime Achievement Award, 2018
- IEEE Outstanding Leadership Award 2018
- UHCL Piper Award Nominee, 2018
- Nominee of USERN Prize 2018
- UHCL Spotlight Researcher, 2018
- Elsevier Outstanding Reviewer, 2018
- IEEE Senior Member, 2016
- ACM Senior Member, 2016
- IEEE Outstanding Leadership Award 2015
- IEEE Outstanding Leadership Award 2014
- Recognition of Workshop Co-Chair from ICCCN 2013
- Distinguished Student Mentor of Meinders School of Business, OCU, 2012
- Fellow of the Faculty Teaching Academy at OCU, 2011
- Semi-Finalist in 2012 Oklahoma Governor’s Business Plan Competition (as advisor), 2012
- The 3rd Place in 2011 Oklahoma Governor’s Business Plan Competition (as advisor), 2011
- Cambridge Who’s Who, 2008
- Marquis Who's Who in America, 2008
- General Motors Academic Award, 2008
- Meritorious Prize of International Undergraduate Mathematical Contest on Modeling (MCM), 2000
- First-Rank Award of Mathematics Contest in Modeling (MCM) of China, 1999
- Baogang Fellowship, by Shanghai Ministry of Education, 2000

Awarded Funding

- NSF CISE-MSI, DP: CNS: An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments, 03/2025-09/2026, \$187,666, PI, Award # 2505686
- NSF CISE MSI Research Expansion, Conference: NSF Computer and Information Science and Engineering Research Expansion Program (CISE MSI) 2024-2026 Aspiring Principal Investigators Workshop, 08/01/2024-07/31/2026, \$100,000, PI, Award # 2437481
- UNT Global Venture Fund (GVF), 11/30/2023-8/10/2024, \$4,000, PI, awarded
- Total external before moving to UNT: **\$5,379,676** (at UHCL); \$20,500 (at OCU)
- NSF CISE-MSI, An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments, 10/2022-09/2025, \$234,982, PI, Award #2240513
- CyBR-MSI 2022 Mini Grant, An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environment, 05/2022-09/2022, \$10,000, PI, awarded
- UH ACPV Grant, 06/2022-09/2022, \$100,000, Senior Personnel (UHCL PI: Dr. Kwok-Bun Yue), awarded
- NASA STTR: Blockchain for Model-Based System Engineering, 2021-2023, Senior Personnel (PI: Dr. Kwok-Bun Yue), \$750,000, awarded
- NSF Noyce: Recruiting and Supporting a Diverse Pool of Highly-Qualified STEM Teachers through Communities of Practice and Informal Science Internships, 04/01/2022-03/31/2026, \$1,199,969, co-PI (PI: Dr. Jana Willis), Award #2151067
- UHCL: Anthony & Barbara Lekkos Endowment Faculty Fellowship, 2021-2022, \$10,000, PI, awarded

- NSF HSI: Building Capacity: Improving STEM Graduation Rates through Engaged Learning, 10/2019-09/2024, co-PI (PI: Dr. Said Bettayeb, other Co-PI: Dr. Hakduran Koc, and Dr. Richard Puzdrowski), \$2,499,930.00, Award #1928622
- NSF DGE: Revamping a Traditional Computer Science Program to Meet Modern-day Cybersecurity Education Challenges, 09/2017-08/2020, co-PI (PI: Dr. Andrew Yang, other Co-PI: Dr. Wei Wei), \$499,995.00, Award #1723596
- NSF: IIP: Creating Augmented Reality Platform for Treatment of Phobias\Anxiety with ASD Children, sub-Award from Baylor College of Medicine, \$4,3800, awarded
- NSF Travel Grant, 2017, PI \$10,000, awarded
- NSF Travel Grant, 2017, PI \$15,000, awarded
- Travel Award from the Sustainable Horizons Institute & Berkeley Lab, 2017, PI, \$1200, awarded
- UHCL Faculty Research Support Funding (FRSF), 2022-2023, PI, \$6,000, awarded UHCL
- Faculty Development Funding (FDF), 2021-2022, PI, \$2,000, awarded
- UHCL Faculty Research Support Funding (FRSF), 2019-2020, PI, \$5,900, awarded
- UHCL Faculty Development Funding (FDF), 2018-2019, PI, \$2,000, awarded
- UHCL Faculty Development Funding (FDF), 2017-2018, PI, \$2,000, awarded
- UHCL Faculty Development Funding (FDF), 2016-2017, PI, \$2,000, awarded
- UHCL Faculty Research Support Funding (FRSF), 2016-2017, PI, \$3,900, awarded
- UHCL Faculty Development Funding (FDF), 2015-2016, PI, \$1,800, awarded
- NSF Travel Grant, 2015, PI \$10,000, awarded
- NSF EPSCoR Oklahoma ROA+, 2015, PI \$12,500 awarded
- OCU Summer Undergraduate Research Grant, 2014, PI \$5000, awarded
- OCU Student Professional Fund, 2014-2015 PI \$1600, awarded
- SC'13 HPC Educator Travel Award, 2013, \$2000, awarded

Publications

Book & Book Chapters

1. **Kewei Sha**, Komala Subramanyam Cherukuri, Haihua Chen, "Privacy and Bias in Edge Computing," in *Edge Computing: Systems and Applications* , IEEE, 2025, pp.223-252, doi: 10.1002/9781394285860.ch8.
2. Haihua Chen, **Kewei Sha**, Yuhua Zhou, and Fengjiao Tu, "Data Quality: Models, Measures, and Tools", *Data Quality, Data Lineage, Data Governance*, Springer, accepted, 2024
3. Michelle Lynn Peters, T. Andrew Yang, Wei Wei, **Kewei Sha**, and Sadegh Davari, "Examining the Influence of Participating in a Cyber Defense Track on Students' Cybersecurity Knowledge, Awareness, and Career Choices", *Advances in Software Engineering, Education, and e-Learning*, pp 31-42, Springer, Cham, 2021
4. **Kewei Sha**, Aaron Striegel and Min Song, "Advances in Computer Communications and Networks -- from green, mobile, pervasive networking to big data computing", edited book, *River Publisher*, 2017
5. **Kewei Sha**, Aaron Striegel and Min Song, "Security, Privacy and Reliability in Computer Communication and Networks", edited book, *River Publisher*, 2017
6. Wenjia Li, **Kewei Sha** and Sherali Zeadally, "Routing in Wireless Sensor Network based Cyber-Physical Systems", book chapter in *Cyber-Physical System Design with Sensor Networking Technologies*, March 2016
7. **Kewei Sha**, "Data Quality Management in Wireless Sensor Networks: A Consistency Model Based Approach", *VDM Verlag Dr. Müller*, April 2010

8. **Kewei Sha**, Weisong Shi, and Sivakumar Sellamuthu, “Load Balanced Query Protocols for Wireless Sensor Networks”, book chapter of the book “*Sensor Networks Operations*”, Edited by Shashi Phoha, Thomas F. La Porta and Christopher Griffin, Wiley-IEEE Press, May 2006

Journals Publications

1. **Kewei Sha**, Kwok-Bun Yue, Wei Wei, Yalong Wu, Madhuri Koduru, and Preethi Vuchuru. "ConsortiumSec: Blockchain-based Distributed Security Framework for Consortium Applications." *Distributed Ledger Technologies: Research and Practice* (2024).
2. Hui Sun, Zhiyong Wang, Yiru Chen, Ying Yu, **Kewei Sha**, and Yalong Wu, “EdgeBrain: A Game based Collaborative Computational Task Offloading Framework for Edge Video Analytics”, *IEEE Transactions on Service Computing (IEEE TSC)* 5, pp. 2287-2303, Sept.-Oct. 2024, doi: 10.1109/TSC.2023.3342800. **(Impact Factor: 5.5, Q1)**
3. Hui Sun, Xiuye Zhang, Bo Zhang, **Kewei Sha**, and Weisong Shi, “Optimal Task Offloading and Trajectory Planning Algorithms for Collaborative Video Analytics with UAV-assisted Edge in Disaster Rescue”, *IEEE Transactions on Vehicular Technology (IEEE TVT)*, vol. 73, no. 5, pp. 6811-6828, May 2024, doi: 10.1109/TVT.2023.3344281 **(Impact Factor: 6.1, Q1) (ABDC A Journal)**
4. Ran Ding, Yan Xu, Hong Zhong, Jie Cui, Jianfeng Ma, **Kewei Sha**, “Towards Fully Anonymous Integrity Checking and Reliability Authentication for Cloud Data Sharing”, *IEEE Transactions on Services Computing (IEEE TSC)*, vol. 16, no. 5, pp. 3782-3795, Sept.-Oct. 2023, doi: 10.1109/TSC.2023.3270580. **(Impact Factor: 5.5, Q1)**
5. Hui Sun, Yiru Chen, **Kewei Sha**, Shaoyuan Huang, Xiaofei Wang, and Weisong Shi, “A Proactive On-Demand Content Placement Strategy in Edge Intelligent Gateways”, *IEEE Transactions on Parallel and Distributed Systems*, vol. 34, no. 7, pp. 2072-2090, July 2023, doi: 10.1109/TPDS.2023.3249797. **(Impact Factor: 5.6, Q1)**
6. Hui Sun, Qiyuan Li, **Kewei Sha**, Ying Yu, “ElasticEdge: Edge Computing-Enabled Intelligent Video Analytics Framework for Live Video”, *IEEE IoT Journal*, vol. 9, no. 22, pp. 23031-23046, 15 Nov.15, 2022, doi: 10.1109/JIOT.2022.3187621. **(Impact Factor: 10.6, Q1)**
7. Hui Sun, Ying Yu, **Kewei Sha***, Hong Zhong, “EdgeEye: A Data-driven Approach for Optimal Deployment of Edge Video Analytics”, *IEEE IoT Journal*, vol. 9, no. 19, pp. 19273-19295, 1 Oct. 1, 2022, DOI: 10.1109/JIOT.2022.3166896. **(Impact Factor: 10.238, Q1)**
8. Sai Ram Vallam Sudhakar, Namarata Kayastha, **Kewei Sha***, “OFES: Optimal Feature Evaluation and Selection for Multi-Class Classification”, *Data & Knowledge Engineering*, 139, 102007, 2022 **(Impact Factor: 1.500, Q2) (ABDC A Journal)**
9. Hui Sun, Bo Zhang, Ying Yu, Xiuye Zhang, **Kewei Sha**, Weisong Shi, “FlexEdge: Dynamic Task Scheduling for a UAV-based On-Demand Mobile Edge Server”, *IEEE IoT Journal*, vol. 9, no. 17, pp. 15983-16005, 1 Sept.1, 2022, doi: 10.1109/JIOT.2022.3152447. **(Impact Factor: 10.238, Q1)**
10. Otmane Azeroual, Meena Jha, Anastasija Nikiforova, **Kewei Sha**, Mohammad Alsmirat, Sanjay Jha, “A Record Linkage-Based Deduplication of Research Data: A DataCleaner Extension”, *Multimodal Technologies and Interaction (Q2)*, 2022; 6(4):27
11. Youhuizi Li, Yuyu Yin, Xu Chen, Jian Wan, Gangyong Jia, **Kewei Sha**, “A Secure Dynamic Mix Zone Pseudonym Changing Scheme Based on Traffic Context Prediction”, *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 9492-9505, July 2022, doi: 10.1109/TITS.2021.3125744. **(Impact Factor: 9.551, Q1) (ABDC A Journal)**
12. Jiaying Tang, Yang Chen, Guozhen She, Yang Xu, **Kewei Sha**, Xin Wang, Yi Wang, and Pan Hui, “Identifying Mis-Configured Author Profiles on Google Scholar Using Deep Learning”, *Applied Sciences*, 11(15), p. 6912, 2021 **(Impact Factor: 2.838, Q2)**

13. Sai Ram Vallam Sudhakar, Namarata Kayastha, **Kewei Sha***, “ActID: An Efficient Framework For Activity Sensor-Based User Identification”, *Computers & Security*, Volume 108, September 2021, 102319 (**Impact Factor: 5.015, Q1**) (**ABDC A Journal**)
14. **Kewei Sha***, Wei Wei, T. Andrew Yang, and Sadeqh Davari, “A Survey of Edge Computing Based Designs for IoT Security”, *Digital Communications and Networks*, 6 (2), 195-202, May 2020. (**Impact Factor: 6.348, Q1**)
15. Hui Sun, Ying Yu, **Kewei Sha***, and Bendong Lou, “mVideo: Edge Computing Enabled Mobile Video Processing Systems”, *IEEE Access*, 8, 11615-11623, 2019 (**Impact Factor: 3.476, Q1**)
16. Youhuizi Li, Zeyong Dong, **Kewei Sha**, Congfeng Jiang, Jian Wan, and Yuan Wang, “TMO: Time Domain Outsourcing Attribute-based Encryption Scheme for Data Acquisition in Edge Computing”, *IEEE Access*, 7 (1), 40240-40257, March 2019 (**Impact Factor: 3.476, Q1**)
17. **Kewei Sha**, Wei Wei, T. Andrew Yang, Zhiwei Wang and Weisong Shi, “Challenges and Open Security Issues in Internet of Thing”, *Future Generation Computer Systems (FGCS)*, 83 (2018), 326-337, June 2018 (**Impact Factor: 5.768, Q1**)
18. **Kewei Sha***, Naif Alatrash and Zhiwei Wang, “A Secure Framework to Read Isolated Smart Grid Devices”, *IEEE Transactions of Smart Grid (TSG)*, 8(6), 2519 - 2531, Nov. 2017 (**Impact Factor: 8.96, Q1**)
19. Aseel Basheer and **Kewei Sha*** “Cluster-based Quality-Aware Adaptive Data Compression for Streaming Data”, *ACM Journal of Data and Information Quality (JDIQ)*, 9(1), 26-63 (article 2). September 2017 (**Impact Score, 2.89**)
20. Junzhao Du, Sicong Liu, Rui Li, Xue Yang, Hui Liu and **Kewei Sha**, “Energy-efficient Algorithm to Construct the Information Potential Field in WSNs”, *IEEE Sensors Journal*, 17(12), 3822 - 3831, June 2017 (**Impact Factor: 3.301, Q2**)
21. **Kewei Sha** and Sherali Zeadally, “Data Quality in Cyber-Physical Systems”, *ACM Journal on Data and Information Quality*, 6(2-3), article No. 8, July 2015 (**Impact Score, 2.89**)
22. Zhiwei Wang, **Kewei Sha** and Wei Lv, “Slight Homomorphic Signature for Access Controlling in Cloud Computing”, *Wireless Personal Communications (WIRE)*, 73 (1), 51-61, 2013 (**Impact Factor: 1.671**)
23. **Kewei Sha**, Jegnesh Gehlot, and Robert Greve, “Multipath Routing Techniques in Wireless Sensor Networks, A Survey”, *Wireless Personal Communications (WIRE)*, 70 (2), 807-829, 2013 (**Impact Factor: 1.671**)
24. **Kewei Sha**, Shinan Wang and Weisong Shi, “RD4: Role-Differentiated Cooperative Deceptive Data Detection and Filtering in VANETs”, *IEEE Transactions on Vehicular Technology (TVT)*, Vol. 59, No. 3, pp. 1183-1190, March 2010 (**Impact Factor: 5.978**) (**ABDC A Journal**)
25. **Kewei Sha** and Weisong Shi, “Consistency-Driven Data Quality Management in Wireless Sensor Networks”, *Journal of Parallel and Distributed Computing (JPDC)*, Vol. 68, No. 9, pp. 1207-1221, September 2008. (**Impact Factor: 3.734, Q1**)
26. Yong Xi, **Kewei Sha**, Weisong Shi, Loren Schwiebert, and Tao Zhang, “Probabilistic Adaptive Privacy-Preserving Authentication in Vehicular Networks”, *Journal of Computer Science and Technology (JCST)*, Vol. 23, No. 6, pp. 916-928, November 2008. (**Impact Factor: 1.571, Q3**)
27. Junzhao Du, Weisong Shi, and **Kewei Sha**, “Asymmetry-Aware Link Layer Services in Wireless Sensor Networks”, *Journal of Embedded Computing*, Vol 3, Issue 2, pp. 141-154, 2009
28. **Kewei Sha**, Junzhao Du, and Weisong Shi, “WEAR: A Balanced, Fault-Tolerant, Energy-Efficient Routing Protocol for Wireless Sensor Networks”, *International Journal of Sensor Networks*, Vol 1, No. 2, 2006 (**Impact Factor: 1.302**)
29. **Kewei Sha**, Weisong Shi, “Modeling the Lifetime of Wireless Sensor Networks”, *Sensor Letters*, Vol 3, pp. 126-135, 2005

Conferences and Workshops Publications

1. Hanzhi Zhang, Heng Fan, **Kewei Sha**, Yan Huang, Yunhe Feng, "DAM: Dynamic Attention Mask for Long-Context Large Language Model Inference Acceleration", In *Findings of the Association for Computational Linguistics: ACL 2025 (ACL 2025)*, Vienna, Austria, July 2025.

2. Hanzhi Zhang, Harold Lucero, **Kewei Sha**, Heng Fan, Song Fu, Yunhe Feng, "D2SO: Detecting Distant and Small Objects for Vision-based Vehicle Autonomous Systems", In *2025 The IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2025)*, Newark, DE, May 2025.
3. Komala Subramanyam Cherukuri, **Kewei Sha**, and Junhua Ding. "Enabling Federated Learning for Object Detection in Connected Autonomous Driving Using YOLO with the Flower Framework" In *34th International Conference on Computer Communications and Networks (ICCCN)*, Tokyo, Japan, August 2025.
4. Yuhan Zhou, Fengjiao Tu, **Kewei Sha**, Junhua Ding, and Haihua Chen. "A Survey on Data Quality Dimensions and Tools for Machine Learning Invited Paper", In *2024 IEEE International Conference on Artificial Intelligence Testing (AITest)*, pp. 120-131. IEEE, 2024.
5. Yalong Wu, Shalini Dhamodharan, Vinuthna Ghattamaneni, Narmada Kokila, Chandrika Pathakamuri, Timothy Carter, Pu Tian, and **Kewei Sha**. "Big Data Quality Scoring for Structured Data Using MapReduce." In *33rd International Conference on Computer Communications and Networks (ICCCN)*, pp. 1-6. IEEE, 2024.
6. Zachary Neal and **Kewei Sha**, "Analysis of Evil Twin, Deauthentication, and Disassociation Attacks on Wi-Fi Cameras", in *Proceedings of the 32th International Conference on Computer Communications and Networks (ICCCN 2023)*, July 2023, Honolulu, HI
7. Otmane Azeroual, Anastasija Nikiforova, and **Kewei Sha**, "The overseen parts of data governance: A workflow framework for dealing with duplicated enterprise data", in *Proceedings of the International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS 2023)*, JUNE 19-22, 2023 - VALENCIA, SPAIN
8. Kwok-Bun Yue, **Kewei Sha**, et. al, Confidentiality and Data Integrity in Consortium Blockchain Applications for Model-Based Systems Engineering, in *Proceedings of the 2023 AIAA SciTech Forum*, Jan 2023, National Harbor, MD
9. Thomas Neumann, Kyle Welsh, Caden Perez, **Kewei Sha**, Yalong Wu, "R²P³: Renewal-Reward Process-Based Password Protection in Internet of Things", in *Proceedings of the 31st International Conference on Computer Communications and Networks (ICCCN 2022)*, Aug 2022, Honolulu, HI
10. Kwok-Bun Yue, Pavani Kallemputi, **Kewei Sha**, Wei Wei, and Xinying Liu. "Governance Attributes of Consortium Blockchain Applications." in *Proceedings of AMCIS 2021*, August 2021.
11. **Kewei Sha**, Wei Wei, Arpit Jain, "Design of Network Forensics Labs for Teaching-oriented Institutions", in *Proceedings of the 2021 International Conference on Computational Science and Computational Intelligence*, Dec 2021, Las Vegas, NV
12. **Kewei Sha**, "Lessons and Experiences From Teaching Computing Science Capstone Project Courses", in *Proceedings of The 2021 International Conference on Computational Science and Computational Intelligence*, Dec 2021, Las Vegas, NV
13. Kwok-Bun Yue, Mark Guerra, Howard Wagner, Joses Sandeep Thamarai Selvan, Kayaanoosh Collector, Victor Tang, Margo Sikes, **Kewei Sha**, Pavani Kallemputi, Sahar Mardani, and Keshav Kasichainula, "Applying Blockchain Technology on Model-Based Systems Engineering", in *Proceedings of the 2021 AIAA SciTech Forum*, January 2021.
14. Jie Cui, Sheng Zhou, Hong Zhong, Yan Xu and **Kewei Sha**, "Transaction-based Flow Rule Conflict Detection and Resolution in SDN", in *Proceedings of the 27th International Conference on Computer Communications and Networks (ICCCN 2018)*, Aug 2018, Hangzhou, China.
15. Hong Zhong, Bo Huang, Jie Cui, Jiang Li and **Kewei Sha**, "Efficient Conditional Privacy-Preserving Authentication Scheme Using Revocation Messages for VANET", in *Proceedings of the 27th International Conference on Computer Communications and Networks (ICCCN 2018)*, Aug 2018, Hangzhou, China.
16. Siguang Chen, Zhihao Wang, and **Kewei Sha**, "Cluster-Aware Kronecker-Supported Data Collection for Sensory Data", in *Proceedings of the 27th International Conference on Computer Communications and Networks (ICCCN 2018)*, Aug 2018, Hangzhou, China.
17. Wei Wei, T. Andrew Yang, Sadegh Davari, **Kewei Sha**, Johanna Jacob, "Toward CAE-CDE 4Y Designation through Curriculum Modernization of a Traditional Computer Science Undergraduate Program", the *Proceedings of EDSIG 2018 conference*, Nov. 2018, Norfolk, VA.
18. Johanna Jacob, Wei Wei, **Kewei Sha**, Sadegh Davari, T. Andrew Yang, "Is the NICE Cybersecurity Workforce Framework (NCWF) Effective for a Workforce Comprising of

- Interdisciplinary Majors?”, in *Proceedings of the 16th International Conference on Scientific Computing (CSC 2018)*, Aug 2018, Las Vegas, NV.
19. Yunxiang Zhang, Xiaokun Yang, Lei Wu, **Kewei Sha**, Archit Manishbhai Gajjar and Han He, “Exploring Slice-Energy Saving on An Video Processing FPGA Platform with Approximate Computing”, in *Proceedings of 2nd International Conference on Algorithms, Computing and Systems*, July 2018, Beijing, China.
 20. Jie Cui, Wenyu Xu, **Kewei Sha** and Hong Zhong, “An Efficient Identity-Based Privacy-Preserving Authentication Scheme for VANETs”, in *Proceedings of 13th EAI International Conference on Collaborative Computing: Networking, Applications and Worksharing (CollaborateCom 2017)*, Dec 2017, EDINBURGH, GREAT BRITAIN.
 21. Mousa Alramadhan and **Kewei Sha**, “An Overview of Access Control Mechanisms for Internet of Things”, in *Proceedings of the 26th International Conference on Computer Communications and Networks (ICCCN 2017)*, Oct 2017, Vancouver, Canada.
 22. Ranadheer Errabelly, **Kewei Sha**, Wei Wei, T. Andrew Yang, Zhiwei Wang, “EdgeSec: Design of an Edge Layer Security Service to Enhance Internet of Things Security”, in *Proceedings of the 1st IEEE International Conference on Fog and Edge Computing (ICFEC 2017)*, May 2017, Madrid, Spain.
 23. **Kewei Sha**, Wei Wei, Andrew Yang and Weisong Shi, “Security in Internet of Things: Opportunities and Challenges”, in *Proceedings of the International Conference on Identification, Information & Knowledge in the Internet of Things (IIKI 2016)*, October 2016, Beijing, China.
 24. Aseel Basheer and **Kewei Sha**, “QAAC: Quality-Assure Adaptive Data Compression for Sensor Data”, in *Proceedings of the 25th International Conference on Computer Communications and Networks (ICCCN 2016)*, August 2016, Waikoloa, Hawaii, USA.
 25. Sicong Liu, Junzhao Du, Rui Li, Hui Liu and **Kewei Sha**, “CrowdBlueNet: Maximizing Crowd Information Collection base on Bluetooth Ad Hoc Networks”, in *Proceedings of The 11th International Conference on Wireless Algorithms, Systems, and Applications (WASA 2016)*, August 2016, Bozeman, Montana, USA.
 26. Wei Wei, Arti Mann, **Kewei Sha** and Andrew Yang, “Design and Implementation of a Multi-Facet Hierarchical Cybersecurity Education Framework”, in *Proceedings of 2016 IEEE International Conference on Intelligence and Security Informatics, the Cybersecurity Workforce and Education track (ISI 2016)*, September 2016, Tucson, USA.
 27. Tom Darling and **Kewei Sha**, “REMEDY: Remote Exercise Monitoring using an Evaluative Data sensing sYstem”, in *Proceedings of 17th International Conference on E-health Networking, Application & Services (IEEE Healthcom 2015)*, October 2015, Boston, USA
 28. Junzhao Du, Sicong Liu, Chi Xu, Kai Wang, Hui Liu and **Kewei Sha**, “Lightweight Construction of the Information Potential Field in Wireless Sensor Networks”, in *Proceedings of the 23rd International Conference on Computer Communications and Networks (ICCCN 2014)*, August 2014, Shanghai, China
 29. **Kewei Sha**, Chengguang Xu and Zhiwei Wang, “One-time Symmetric Key Based Cloud Supported Secure Smart Meter Reading”, in *Proceedings of the 23rd International Conference on Computer Communications and Networks (ICCCN 2014)*, August 2014, Shanghai, China
 30. Myung Ah Park, Gang Qian, Michelle Hepner, **Kewei Sha**, “A Gender-Inclusive Holistic approach for IAS Education”, in *Proceedings of the International Conference on Frontiers in Education: Computer Science and Computer Engineering*, July 2012, Las Vegas, Nevada, USA
 31. Chang Liu, Junzhao Du, Hui Liu and **Kewei Sha**, “Sleep-Wakeup Algorithms for Virtual Barriers for Wireless Sensor Networks in 3D Space”, in *Proceedings of the 6th International Conference on Mobile Ad-hoc and Sensor Networks (MSN 2010)*, Dec 2010, Hangzhou, China
 32. Yawei Li, Junzhao Du, Hui Liu and **Kewei Sha**, “On Sweep Coverage with Minimum Mobile Sensors”, in *Proceedings of the 16th International Conference on Parallel and Distributed Systems (ICPADS 2010)*, Dec 2010, Shanghai, China
 33. **Kewei Sha**, Guoxing Zhan, Safwan Al-Omari, Tim Calappi, Weisong Shi and Carol Miller, “Data Quality and Failures Characterization of Sensing Data in Environmental Applications”, In *Proceedings of CollaborateCom 2008*, November 2008.
 34. **Kewei Sha**, Guoxing Zhan, Weisong Shi, Mark Lumley, Clairi Wiholm, and Bengt Arnetz, “SPA: Smart Phone Assisted Chronic Illness Self-Management with Participatory Sensing”, in *Proceedings of The 2nd International Workshop on Systems and Networking Support for*

- Healthcare and Assisted Living Environments (Healthnet'08)*, in conjunction with MobiSys'08, June, 2008.
35. **Kewei Sha** and Weisong Shi, "Modeling Data Consistency in Wireless Sensor Networks" in *Workshop Proceedings of ICDCS 2007 (WWASN 2007)*, Toronto, June 25-29, 2007
 36. Yong Xi, **Kewei Sha**, Weisong Shi, Loren Schwiebert, and Tao Zhang, "Enforcing Privacy Using Symmetric Random Key-Set in Vehicular Networks" in *Proceedings of the 8th International Symposium on Autonomous Decentralized Systems (ISADS)*, Sedona AZ, March, 2007
 37. **Kewei Sha**, Yong Xi, Weisong Shi, Loren Schwiebert, and Tao Zhang, "Adaptive Privacy-Preserving Authentication in Vehicular Networks", in *Proceedings of the First International Conference on Communications and Networking in China (ChinaCom'06)*, Oct 27, 2006
 38. **Kewei Sha** and Weisong Shi, "On the Effects of Consistency in Data Operations in Wireless Sensor Networks", in *Proceedings of the 2006 International Conference on Parallel and Distributed Systems (ICPADS 2006)*, July 12-15, 2006
 39. **Kewei Sha**, Weisong Shi, and Orlando Watkins, "Using Wireless Sensor Networks for Fire Rescue Applications: Requirements and Challenges", in *Proceedings of the 6th IEEE International Conference on Electro/Information Technology*, May 2006
 40. Junzhao Du, Weisong Shi, and **Kewei Sha**, "Asymmetry-Aware Link Quality Services in Wireless Sensor Networks", in *Proceedings of the 2005 IFIP International Conference on Embedded and Ubiquitous Computing (EUC-05)*, December 6-9, 2005
 41. Weisong Shi, Sivakumar Sellamuthu, **Kewei Sha**, and Loren Schwiebert, "QueryAgent: A General Query Processing Tool for Sensor Networks", in *Workshop Proceedings of International Conference on Parallel Processing (ICPP'04)*, Aug 2004

Journal Editorial

1. **Kewei Sha**, Xiaohui Liang and Wenjia Li "Editorial: Special Issue on Security in Medical Cyber-Physical Systems", Elsevier Smart Health, November 2018.
2. Shiwen Mao, Honggang Wang, Aaron Striegel, and **Kewei Sha**, "Editorial: Special Issue on Cognitive, Cellular, and Mobile Networks", *EAI Endorsed Transactions on Wireless Spectrum*, 16(7): e1, January 2016.
3. Abhishek Parakh and **Kewei Sha**, "Editorial: Privacy, Security and Trust in Mobile and Wireless Systems", *International Journal of Security and Networks*, Vol. 11, Nos. 1/2, 2016
4. **Kewei Sha** and Zhengping Wu, "Editorial: Privacy, Security and Trust in Mobile Systems", *Wireless Personal Communications*, 73(1), 1-3, 2013

Poster Papers

5. Sina Montazeri, **Kewei Sha**, Walteneagus Dargie, and Yunhe Feng, "Activity Recognition based on ECG only", in *2025 Texas Health Informatics Alliance Conference*, September 26, Arlington, TX.
6. Sina Montazeri, **Kewei Sha**, Walteneagus Dargie, and Yunhe Feng, "Evaluating Federated Object Detection for Connected Autonomous Vehicles", in *2025 InfoFORCE: Information and AI Weekend*, September, Frisco, TX.
7. Sina Montazeri, **Kewei Sha**, Walteneagus Dargie, and Yunhe Feng, "Activity Recognition based on ECG", in *2025 InfoFORCE: Information and AI Weekend*, September, Frisco, TX.
8. Yuhua Zhou, Fengjiao Tu, **Kewei Sha**, Junhua Ding, and Haihua Chen, "A Survey on Data Quality Dimensions and Tools for Machine Learning", in *Day of Health Informatics and Data Science*, September 2024, Frisco, TX. (the **3rd prize** for Data Science student poster competition)
9. Komala Subramanyam Cherukuri, **Kewei Sha**, and Haihua Chen, "Evaluation of Privacy-Preserving Techniques for Edge Computing", in *Day of Health Informatics and Data Science*, September 2024, Frisco, TX. (the **2nd prize** for Data Science student poster competition).
10. Ross Niswanger, **Kewei Sha**, and Dorothea C Lerman, "Design of Mixed Reality Dangerous Situations for Autistic Children: Road Safety", in *Proceedings of the 2023 ACM/IEEE CHASE conference*, June 2023, Orlando, FL.

11. Yalong Wu, **Kewei Sha**, and Kwok-Bun Yue, “Blockchain-Enabled Federated Edge Learning for Big Data Quality Assessment”, in *Proceedings of the 2022 IEEE/ACM SEC conference, Dec 2022, Seattle, WA*
12. Hui Sun, Yiru Chen, **Kewei Sha**, and Yalong Wu, “Ensemble Federated Edge Learning for Recommender Systems”, in *Proceedings of the 2022 IEEE/ACM SEC conference, Dec 2022, Seattle, WA*
13. **Kewei Sha**, Piyush Manghani, and Dorothea C Lerman, “Augmented Reality Game Therapy for Children with Autism Spectrum Disorder”, in *Proceedings of the 2022 ACM/IEEE CHASE conference, Nov 2022, Washington, DC.*
14. **Kewei Sha** and Youhuizi Li, “Design of Chord Ring and SDN Based Edge Infrastructure Management System” (3-page poster summary), in *Proceedings of The Fourth ACM/IEEE Symposium on Edge Computing*, Nov 2019, Washington, DC.
15. **Kewei Sha**, Zhandong Liu, and Jack Dempsey, “Augmented Reality Based Therapy System for Social Skill Deficits” (2-page poster summary), in *Proceedings of The Fourth IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2019)*, Sep 2019, Washington, DC.
16. Namrata Kayastha and **Kewei Sha**, “A Novel and Efficient Approach to Evaluate Biometric Features for User Identification” (2-page poster summary), in *Proceedings of The Fourth IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2019)*, Sep 2019, Washington, DC.
17. **Kewei Sha** and Madhu Kumari, “Patient Identification Based On Wrist Activity Data”, in *Proceedings of The Third IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2018)*, Sep 2018, Washington, DC.
18. **Kewei Sha**, “EdgeSec: Edge Computing for IoT Security”, in *Proceedings of Sustainable Research Pathways Workshop* organized by SHI Sustainable Research Pathways Institute 2017, Dec 2017, Lawrence Berkeley National Laboratory, CA.
19. Naveen Manoharan, **Kewei Sha**, and Jiang Lu, “Exercise Evaluation Using Wrist Sensors”, in *Proceedings of The Second IEEE/ACM Conference on Connected Health: Applications, Systems, and Engineering Technologies (IEEE/ACM CHASE 2017)*, July 2017, Philadelphia, PA.
20. Nitish Kelagote, **Kewei Sha**, Jiang Lu, Lin Sun and Sanobar Kadiwal, “Human Motion Modeling and Evaluation Using Wearable Sensors”, in *Proceedings of the Dual Conference 2016 in Innovation and Automation*, October 2016, Houston, USA.
21. Sanobar Kadiwal, Jiang Lu, **Kewei Sha**, Lin Sun and Nitish Kelagote, “Microsoft Kinect Based Real-time Segmentation and Recognition for Human Activity Learning”, in *Proceedings of the Dual Conference 2016 in Innovation and Automation*, October 2016, Houston, USA.
22. Rohit Reddy, **Kewei Sha**, Jegnesh Gehlot and Robert Greve, “On the Heterogeneity in Mobile Application Development”, in *Proceedings of Oklahoma Research Day*, Nov 2011, Lawton, Oklahoma, USA
23. Shinan Wang, **Kewei Sha** and Weisong Shi, “Role-based Deceptive Data Detection and Filtering in WSNs”, in *Proceedings of the 8th ACM/IEEE International Conference on Information Processing in Sensor Networks (IPSN 2009)* (2 page summary), April 13-16, 2009, San/Francisco.
24. **Kewei Sha** and Weisong Shi, “Revisiting the Lifetime of Wireless Sensor Networks”, in *Proceedings of the 2nd International Conference on Embedded Networked Sensor Systems (SenSys’04)* (poster), November 3-5, 2004

Supervised Master Thesis

1. Parth Trada, “Evaluating Sentiment Analysis Mechanisms For Amazon Review”, May 2023
2. Ritu Kadve, “Federated Learning To Build Sentiment Analysis Models for Amazon Review Dataset Without Labels”, May 2023
3. Priyanka Kumari, “Tagging Negative Review in Amazon Product Review”, May 2023
4. Sai Ram Vallam Sudhakar, “An Efficient Classification Algorithm for Gait Authentication based on Streaming Sensory Data”, May 2021

5. Namarata Kayastha, "Biometrics-based User Identification with Optimal Feature Evaluation and Selection", master thesis, Dec 2019
6. Mahanad Ramadhan, "*EBMR: Energy-Efficient and Load-Balanced Multipath Routing Protocol for Wireless Sensor Networks*", master thesis, May 2016
7. Aseel Basheer, "*QAAC: Quality-Assured Adaptive Data Compression for Sensor Data*", master thesis, Dec 2015

Recent Talks

- *Enabling Federated Learning for Object Detection in Connected Autonomous Driving Using YOLO with the Flower Framework*, 34th International Conference on Computer Communications and Networks (ICCCN), Tokyo, Japan, August 2025.
- *Big Data Quality Scoring for Structured Data Using MapReduce*, 33rd International Conference on Computer Communications and Networks (ICCCN), Big Island, HI, July 2024
- *Towards Efficient, Secure, and Quality-aware Data Analytics*, **Keynote** talk at the 4th Interdisciplinary Conference on Electrics and Computer, Chicago, IL, June 2024
- *An Edge-Based Approach to Robust Multi-Robot Systems in Dynamic Environments*, poster presentation, 2024 CISE Research Expansion Grantee Meeting, Denver, CO, May 2024
- *Effective Collaborations among Six Institutions*, 2024 CISE Research Expansion Grantee Meeting, Denver, CO, May 2024
- *Edge-computing Supported Robust Multi-Robot Systems*, talk at Information Science Brown Bag Research Meeting at University of North Texas, Sep 2023
- *User Identification, Authentication, and Secure Storage*, Invited talk at University of Houston, March 2023
- *ActID: Activity Sensor Based User Identification*, Invited virtual talk at Florida Gulf Coast University, Jan 2023
- *Efficient, Secure, and Privacy-preserving Data Analytics and Sharing*, Invited talk at University of North Texas, Dec 2022
- *Edge Computing Designs for IoT Security*, talk at HSI Summer Camp, July 15, 2022
- *How to write research papers*, invited virtual talk at Anhui University, November, 2021
- *Cybersecurity Research at UHCL*, invited talk at UHCL, October, 2021
- *ActID: An Efficient Framework for Activity Sensor Based User Identification*, Invited virtual talk at Nanjing University of Posts and Communications, May 2021
- *Data Quality Management in IoT*, invited virtual talk at Anhui University, November, 2020
- *Edge Computing Based Designs for Security and Data Analytics*, invited talk at North Carolina Central University, April 28, 2020
- *Edge Computing Based Designs for Security, Performance Optimization, and Data Analytics*, invited talk at Oakland University, January 23, 2020
- *Edge Computing based Security Designs*, invited talk at SIG of InfraGard Defense Industrial Base, Jan 14, 2020
- *An Introduction to NICE Cybersecurity Workforce Framework*, talk at Cybersecurity Education Workshop, April 5, 2019
- *The Design of Network Forensics Course*, talk at Cybersecurity Education Workshop, April 5, 2019
- *Studies on Edge Computing for IoT Security*, invited seminar talk at Wayne State University, March 27, 2019
- *Research on Edge Computing Security*, invited talk at Anhui University, August 2018
- *Home-based Exercise Evaluation Using Wrist Sensors*, Keynote at Healthcare IT 2018 at World Congress & Expo on Healthcare IT, August 2018

- *Security Issues in Edge Computing*, Colloquium talk at University of Houston – Clear Lake, April 23, 2018
- *Introduction to Internet of Things*, Invited talk at Math Center at University of Houston – Clear Lake, September 2017
- *Quality-Assured Adaptive Data Compression for Sensor Data*, Invited talk at Nanjing University of Posts and Communications, June 2017
- *Emerging Cyber Threat: Challenges & Opportunities*, Keynote at University of Houston System Cybersecurity Cyber-Resilience Workshop, March 2017
- *IoT Security: challenges and solutions*, invited talk at Zhejiang Wanli University, January 2017
- *Security Issues in Internet of Things: opportunities and challenges*, invited talk at Ningbo University, January 2017
- *IoT Security: opportunities and solutions*, invited talk at Shanghai University of International Business and Economics, January 2017
- *Security in Internet of Things: vision and challenges*, Invited talk at Anhui University, December 2016
- *Edge Supported Security Services for Internet of Things*, Invited talk at Nanjing University of Posts and Communications, December 2016
- *IoT Security: vision, challenges, and solutions*, Keynote at International Conference on Wireless Communication and Sensor Network (WCSN 2016), December 2016
- *The World of IoT*, invited talk at The Chinese Entrepreneur Association, April 2016
- *IoT Security: Perspectives, Challenges, and Cases*, invited Round Table talk at University of Nebraska at Omaha, April 2016
- *Energy Efficient Data Quality Management in Wireless Sensor Systems*, invited talk at University of Houston – Clear Lake, May 2015
- *A Secure Framework to Read Isolated Devices in Smart Grid*, invited talk at Howard University, Apr 2015
- *Data Quality Management in WSN*, invited talk at Eastern Washington University, Apr 2015
- *Role-based Deceptive Data Detection in VANET*, invited talk at University of Indianapolis, Feb 2015
- *A Cloud-aided Framework for Securely Reading Isolated Devices in Smart Grid*, invited talk at Nanjing University of Posts and Communications, July 2014
- *An Efficient Framework for Data Quality Management*, invited talk at Nanjing University of Posts and Communications, July 2014
- *Data Quality in Wireless Sensor Networks*, invited talk at Oklahoma City University, Oct 2013

Panelist

- *NSF proposal review procedures and tips*, 2025 NSF CISE MSI Aspiring PI Virtual Workshop, January 2025
- *Preparing Budget for NSF proposals*, 2025 NSF CISE MSI Aspiring PI Virtual Workshop, January 2025
- *Preparing Competitive NSF Proposals of Engineering and Computing Education*, NSF Proposal Writing Workshop for Hispanic-Serving Institution, 2023 IEEE ASEE Frontiers in Education Conference, College Station, Texas, October 18, 2023
- *Security Education and Research at UNT*, Cyber Awareness Month Series – Panel Discussion, University of North Texas, September 15, 2023
- *Doing Research at UHCL*, Faculty Development Day at University of Houston – Clear Lake, April 6, 2018
- *Security Education and Research at UHCL*, University of Houston System Cybersecurity Cyber-Resilience Workshop, March 10, 2017

Professional and Scholarly Participation

Funding Proposal Reviewer

- NSF Panelist, 2025
- Hong Kong Research Grants Council, 2024
- Oakland University Research Grant, 2024
- NSF Panelist, 2021
- NSF Panelist, 2020
- NSF Panelist, 2017
- Sam Houston State University Research Fund, 2017
- NSF Bluewaters Scholarship, 2014, 2015, 2016, 2017, 2018
- University of Central Oklahoma On-campus Interdisciplinary Grant, 2014
- University of Pisa Research Proposal 2008

Professional Membership

- Senior Member of IEEE
- Senior Member of ACM

Professional Services

- **Editorship**
 - **Associate Editor**
 - Smart Health (Elsevier), 2019 -
 - Computing (Springer), 2020 -
 - Multimedia Tools and Applications (Springer), 2023 -
 - Frontiers in Communications and Networks (Frontier), 2025-
 - Frontiers in Internet of Things (Frontier), 2025-
 - **Editorial Board Member**
 - IEEE IoT Journal, 2021 -
 - Computer Systems Science and Engineering, 2023 -
 - **Guest Editor**
 - Special Issue: Advances in Generative AI and Computational Linguistics, Electronics, 2025-2026
 - Special Issue in Advanced Deep Learning and Sensing Techniques for Complex Structural Health Monitoring, Sensors, 2022-2023
 - Special Issue in New Advances in Securing Data and Big Data, Algorithms, 2021-2022
 - Special Issue in Security and Privacy in Internet of Medical Things (IoMT), Security and Communication Networks, 2019-2020
 - Topic collections in Security and Privacy in Smart Connected Health Systems, Journal of Medical Systems, 2018-2019
 - Special Issue in Security of Medical CPS, Elsevier Smart Health, 2017-2018
 - Special Issue in Privacy, Security and Trust in Mobile Systems, Journal of Security and Network, 2016
 - Special Issue in EAI Endorsed Transactions on Wireless Spectrum, 2016
 - Special Issue in Mobile Privacy, Security and Trust, Personal Wireless Communications, 2013
 - **Member of Reviewer Board**
 - Future Internet, 2022 -
- **Organization Officer/Secretary**
 - **Founding Board of Directors and Secretary of International Autonomous Driving Organization, 2021 - present**

- Executive Committee Member and Communication Director, Association of Chinese Scholars in Computing, 2023 - present
- Secretary of Technical Committee on the Internet of the IEEE Computer Society (IEEE-CS TCI) (2013 - 2016)
- **Conference Organizing Committee & Chair**
 - **Steering Committee Member**
 - IEEE International Conference on Mobility: Operations, Services, and Technologies, 2025 -
 - **Organizing Committee Advisory Member**
 - International Conference on Computer Communications and Networks, 2023-
 - **Conference General Chair**
 - The 2nd IEEE Mobility: Operations, Services, and Technologies (MOST 2024)
 - The 8th ACM/IEEE Symposium on Edge Computing (SEC 2023)
 - The 6th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2021)
 - The Third International Conference on Wireless Communication and Sensor Network (WCSN 2016)
 - **Program co-Chair**
 - NSF CISE Research Expansion (RE) Aspiring PI Workshop at FIU, 2025-2026
 - NSF CISE MSI Aspiring PI Workshop at UNT, 2024-2025
 - The 2nd International Conference on Cybersecurity and Artificial Intelligence Strategies (CAIS'25)
 - The 24th International Conference on Computer Communications and Networks (ICCCN 2015)
 - **Workshop General Chair**
 - The International Conference on Computer Communications and Networks (ICCCN 2018-2026)
 - The 22nd International Conference on Computer Communications and Networks (ICCCN 2013)
 - The 11th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2026)
 - The 10th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2025)
 - The 9th ACM/IEEE Symposium on Edge Computing (SEC 2024)
 - **Symposium Chair**
 - The 2019 International Conference on Computing, Networking and Communications (ICNC 2019)
 - **Track Chair**
 - The 26th International Conference on Computer Communications and Networks (IEEE ICCCN 2017)
 - **Student Travel Grant Chair**
 - The 14th IEEE International Conference on Sensing, Communication and Networking (IEEE SECON 2017)
 - The 2nd IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2017)
 - The 25th International Conference on Computer Communications and Networks (ICCCN 2016)
 - **Publication Chair**
 - The First IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2023)
 - The ACM/IEEE Symposium on Edge Computing (SEC 2020-2022)
 - The IEEE International Conference on Connected and Autonomous Driving (MetroCAD 2020-2022)

- The IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (2017-2020)
 - **Workshop Co-Chair**
 - The Eighth ACM/IEEE Workshop on Security and Privacy in Edge Computing (EdgeSP 2025)
 - The Workshop on Synergy of AI and Multimodal Biomedical Data Mining (co-located with ICDM 2025)
 - The International Workshop on Security, Privacy, and Trustworthiness in Medical Cyber Physical Systems (MedSPT 2016-2018)
 - University of Houston System Cybersecurity Cyber-Resilience Workshop, 2017
 - The International Workshop on Privacy, Security and Trust in Mobile and Wireless Systems (MobiPST 2011, 2012, 2014)
 - **Session Chair**
 - The 8th IEEE Symposium on Edge Computing (SEC 2023)
 - The First IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST 2023)
 - The International Conference on Computer Communications and Networks (ICCCN 2011, 2013-2017, 2020)
 - The 4th IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2019)
 - The 17th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid 2017)
 - The 3rd International Workshop on Context-aware QoS Provisioning and Management for Emerging Networks, Applications and Services (ContextQoS 2013)
 - The International Conference on Collaborative Computing (CollaborateCom 2008-2009)
 - **Publicity Chair**
 - International Conference on Computing, Networking and Communications (ICNC 2017)
 - The 2017 International Conference on Applications and Techniques in Cyber Intelligence (ATCI 2017)
 - 2013 International Workshop on Mobile Cloud Computing Technologies and Applications (MCCTA 2013)
 - The 3rd International Workshop on Privacy, Security and Trust in Mobile and Wireless Systems (MobiPST 2013)
- **TPC member**
 - The IEEE International Conference on Trust, Security and Privacy in Computing and Communications (IEEE TrustCom 2025)
 - 2024 IEEE World Forum on Public Safety Technology (WF-PST)
 - IEEE International Conference On Artificial Intelligence Testing (AITesting 2024)
 - The IEEE International Conference on Communications (IEEE ICC 2016, 2019-2021, 2023-2024)
 - International Conference on Computing, Networking and Communications (ICNC 2023-2024)
 - The 8th IEEE International Conference on Intelligent Data and Security (IEEE IDS 2022)
 - The 15th International Conference on Knowledge Science, Engineering and Management (KSEM 2022)
 - The IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technology (CHASE 2018-2021)
 - The IEEE International Conference on Communications (IEEE ICC 2016, 2019-2021, 2023-2024)

- 2021 4th International Conference on Computer Applications & Information Security (ICCAIS'2021)
- The International Conference on Computer Communications and Networks (ICCCN 2013, 2015-2017, 2020-2021)
- International Conference on Algorithms and Architectures for Parallel Processing (ICA3PP 2020)
- The International Conference on Wireless Algorithms, Systems, and Applications (WASA 2016-2021)
- The 2020 International Conference on Computing, Networking and Communications (ICNC 2020)
- The 25th IEEE International Conference on Parallel and Distributed Systems (ICPADS 2019)
- The 4th IEEE International Conference on Intelligent Data and Security (IEEE IDS 2019)
- The IEEE International Conference on Computational Science and Engineering (IEEE CSE 2012-2013, 2019)
- The International Conference on Green, Pervasive and Cloud Computing (GPC 2013, 2015, 2017-2019)
- National Cyber Summit (NCS 2018-2021)
- The 2nd International Workshop on Attacks and Defenses for Internet-of-Things (ADIoT 2019)
- The IEEE International Conference on Ubiquitous Computing and Communications (IEEE IUCC 2017-2018)
- The IEEE International Conference on Trust, Security and Privacy in Computing and Communications (IEEE TrustCom 2016-2018)
- The 2017 IEEE International Conference on Collaboration and Internet Computing (IEEE CIC 2015-2017)
- The IEEE Conference on Local Computer Networks (IEEE LCN 2016-2017)
- International Conference on Agents and Artificial Intelligence 2017
- The International Workshop on Cloud Storage Service and Computing, 2016-2017
- 2017 IEEE WCNC Workshop on M2M Communications and the Internet of Things
- The 35th IEEE International Performance, Computing, and Communication Conference (IPCCC 2016)
- The annual International Conference on Distributed Computing in Sensor Systems (DCOSS 2016)
- The International Conference on Security of Smart Cities, Industrial Control System and Communications (SSIC 2016)
- The 2015 Global Conference on Information Technology, Computing, and Applications (Global IT 2015)
- The Third International Symposium on Security in Computing and Communications (SSCC'15)
- The Sixth International Conference on Mobile Computing, Applications and Services (MobiCASE 2014)
- The 5th International Conference on Creative Converged IT (CCIT 2014)
- The IEEE International Conference on High Performance Computing and Communications (IEEE HPCC 2013-2014)
- The 2014 FTTRA International Symposium on Ubiquitous Green IT Systems (U-GIT 2014)
- The 2014 IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2014)
- The IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing (CollaborateCom 2009-2014)
- The Workshop on Big Data and e-Health, 2014

- The International Conference on Ambient Systems, Networks and Technologies (ANT), 2013
- The International Workshop on Internet of Ubiquitous and Pervasive Things (IUPT 2011-2014)
- The International Workshop on Data Management for Cyber Physical System (DM-CPS), 2013
- The World Conference on Information Systems and Technologies (WorldCIST) 2013
- The International Conference on Advanced Collaborative Networks, Systems and Applications (COLLA 2011-2014)
- The International Conference on Advanced Engineering Computing and Applications in Sciences (ADVCOMP 2011-2014)
- The International Conference on Mobile Web Information Systems (MobiWIS) 2012
- The International Workshop on Sensor Networks (SN) 2012
- The IEEE International Conference on Scalable Computing and Communications (IEEE ScalCom) 2012
- The IEEE International Conference on Wireless Communications, Networking and Information Security (IEEE WCNIS) 2010
- The IEEE International Workshop on Mobile Cyber-Physical Systems (MobiCPS) 2010
- The IEEE International Conference on Green Computing and Communications (IEEE GreenCom) 2010
- The International Workshop on Wireless Ad hoc and Sensor Networking (WWASN 2011)

- **Journal Reviewer**

- ACM Transactions on Computing for Healthcare, 2025
- Nature Scientific Reports, 2025
- IEEE Internet of Things Journal, 2016- 2022, 2024,2025
- ACM Transactions on Internet of Things, 2021-2022, 2024
- ACM Computing Survey, 2024
- Elsevier Smart Health, 2017-2024
- IEEE Transactions on Emerging Topics in Computing, 2023
- IEEE Transactions on Vehicular Technology, 2016-2022
- IEEE Transactions on Green Communications and Networking, 2021
- IEEE Sensor Journal, 2020
- IEEE Transactions on Smart Grid, 2020
- IEEE Journal of Selected Areas in Communications, 2020
- IEEE Transactions on Information Forensics & Security, 2020
- ACM Transactions on Computing for Healthcare, 2019-2021
- IEEE Internet Computing, 2019, 2020
- IEEE Transactions on Big Data, 2017, 2018, 2020
- IEEE Transactions on Industrial Informatics, 2018, 2019
- IEEE Transactions on Service Computing, 2017- 2019, 2021, 2023
- IEEE Transactions on Network Science and Engineering, 2018, 2020
- IEEE Transactions on Dependable and Secure Computing, 2014, 2020
- IEEE Network Magazine, 2020-2023
- IEEE Journal of Biomedical and Health Informatics, 2023
- Multimedia Tools and Applications, 2023
- International Journal of Machine Learning and Cybernetics, 2020
- Digital Communications and Networks, 2020
- Wireless Communications and Mobile Computing, 2020
- Computers, Materials & Continua, 2020
- Information Systems Frontiers, 2020
- ACM Transactions on Information Technology (TOIT), 2016-2017, 2019-2021
- IEEE ACCESS, 2017-2019
- Transactions on Emerging Telecommunications Technologies, 2019
- IEEE Transactions on Engineering Management, 2018

- IEEE Communications Letters, 2018
- IEEE Transactions on Mobile Computing, 2017, 2018
- IEEE/ACM Transactions on Networks, 2017, 2018
- IEEE Transactions on Parallel and Distributed Systems (TPDS) 2008, 2010-2017
- IEEE Transactions on Cloud Computing, 2017
- IEEE System Journal, 2015, 2018
- Elsevier Smart Health, 2017-2022
- Future Generation Computer Systems, 2018
- Journal of Network and Computer Applications, 2018
- IEEE Reviews in Biomedical Engineering, 2018
- Journal of Parallel and Distributed Computing (JPDC) 2007, 2010, 2011, 2018
- Sustainable Computing, Informatics and Systems, 2017
- EURASIP Journal on Wireless Communications and Networking, 2017, 2018
- IEEE Sensor Journal, 2016
- IEEE Internet Journal, 2016
- Computing (COMP), 2013, 2017, 2018
- International Journal of Communication Systems, 2017
- Security and Communication Networks, 2017, 2022
- Computational Intelligence, 2016
- Sensors, 2016, 2017
- Elsevier's Computer Networks (ComNet) 2011, 2014, 2016, 2017, 2018
- Elsevier's Computer Communications Journal (ComCom) 2008, 2011, 2012, 2013, 2014, 2016, 2020
- Wireless Networks, 2016
- Ad Hoc Networks 2013, 2014, 2015, 2016
- Future Internet, 2022
- IEEE Transactions on Computers, 2013
- IEEE Transactions on Intelligent Transportation Systems (TITS), 2013, 2023
- Transactions on Emerging Telecommunications Technologies, 2024
- International Journal of Security and Networks, 2015, 2016
- International Journal of Ad Hoc and Ubiquitous Computing 2009, 2010, 2013, 2016
- Wireless Personal Communications (WIRE) 2009-2014, 2016
- IET Wireless Sensor Systems, 2015
- ETRI Journal 2014
- Journal of Systems Architecture, 2012, 2013
- Journal of Network and Computer Applications 2013, 2014
- International Journal of Sensor Networks (IJSNet) 2009, 2012-2014, 2016-2017
- Journal of Sensors 2009, 2012, 2014, 2015
- Mobile Networks and Applications (MONE), 2010, 2011, 2013
- International Journal On Advances in Intelligent Systems 2012-2014
- International Journal On Advances in Software 2012-2014
- International Journal of Parallel, Emergent and Distributed Systems, 2012
- IST 2012
- BMC Bioinformatics, 2012
- International Journal of Systems Science, 2010
- International Journal of Distributed Sensor Networks (IJDSN), 2011
- Journal of Computer Networks and Communications (JCNC), 2011
- Mathematical and Computer Modeling (MCM), 2010, 2011
- International Journal of Internet Protocol Technology (IJIPT) 2010
- International Journal of Intelligent Engineering & Systems 2009
- ACM Transactions on Autonomous Adaptive Systems (ACM TAAS) 2008
- Journal of Computer Science and Technology (JCST) 2008, 2010
- Journal of Computers and Applications 2007

- **Other Services**

- Co-Organizer, Data Science Talk Series, 2024, 2025

- UNT Data Science New Student Orientation, 2024
- Mentor, NSF CISE MSI Proposal Writing, 2024, 2025
- Ph.D. dissertation committee member, Information Science, UNT, 2024 – 2025
- Ph.D. dissertation committee member, Computer Science and Engineering, UNT, 2024 -
- Judge, UNT Data Science Poster Competition, 2024
- Judge, 2024 UNT Multi-disciplinary Information Research Symposium (MIRS), 2024
- Judge, UHCL STEM Challenge, 2022
- Ph.D. dissertation committee member, Wayne State University, 2022-
- Sponsor, Clear Lake High School STEM Challenger Team, 2022
- Mentor, programming training for high school students, 2020
- Board Member of Bay Area Youth Symphony, 2021
- Mentor of High School Independent Study in IoT, 2021-2022
- Invited External Reviewer of faculty promotion and tenure, University of Washington Bothell
- Invited External Reviewer of faculty promotion and tenure, Saint Joseph's University
- Invited External Reviewer of faculty promotion and tenure, University of Nebraska–Omaha
- Invited External Reviewer of faculty promotion and tenure, Lewis University
- Invited External Reviewer of faculty promotion and tenure, University of Massachusetts Boston
- Judge of InnoSTARS Competition, 2018
- Judge of FIRST Robotics Competition, Oklahoma Region, 2013, 2014

University Services

New Program Development

- *MS. in Data Science*, University of Houston-Clear Lake
- *Minor in Cybersecurity for Computer Science Students*, University of Houston-Clear Lake

University Systems, University, College and Department Service

- Director, Data Science Graduate Program, 2024 –
- Member, College of Information Curriculum Committee (Fall, 2025 -)
- Member, College of Information, Dean Evaluation Committee (Spring, 2025)
- Member, Data Science Tenured Faculty Search Committee (Spring, 2025)
- Chair, Data Science Curriculum Committee (Spring, 2025 -)
- Chair, Data Science Reappointment, Promotion and Tenure Committee (Spring, 2025 -)
- Chair, Data Science Annual Evaluation Committee (Spring, 2025 -)
- Member, Data Science Workload Committee (Spring, 2025 -)
- Co-Chair, Department of Information Science Reappointment, Promotion and Tenure Committee (Fall, 2024)
- Chair, Data Science Curriculum Work Group (Fall, 2024)
- Member, Department of Information Science Faculty Personnel Affairs Committee (Fall, 2024)
- Mentor, Junior Faculty Mentorship (Fall, 2024)
- Member, Data Science Tenure-track Faculty Search Committee for three tenure track assistant professors (Spring, 2024)
- Member, Data Science Faculty Search Committee at UNT, 2023-2024
- Member, External Affairs & Physical Resources Committee at UNT, 2023-2024
- Member, UH Systems Vice Chancellor/Vice President for Research Search Committee, 2023
- Faculty Advisor, UHCL ACM Chapter, 2021 - 2023
- Member, UHCL President's outstanding faculty award committee, 2020
- Member, College of Science and Engineering (CSE) FDF Committee at UHCL, 2021- 2023

- Member, CSE Mentoring Committee at UHCL 2021- 2023
- Member, CSE Graduate Research Assistant Committee at UHCL, 2017-2018
- Member, CSE Committee of Computing, Library, Research Services at UHCL, 2019-2021
- Member, CSE Policy and Advisory Committee at UHCL, 2019-2021
- Member, Computer Science (CS) New Faculty Search Committee at UHCL, 2022
- Member, Data Science (DS) New Faculty Search Committee at UHCL, 2022
- Member, CS Promotion and Tenure Review Committee at UHCL, 2021
- Member, Computer Engineering (CE) Promotion and Tenure Review Committee at UHCL, 2020
- Member, DS Committee at UHCL, 2020- 2023
- Member, DS Admission Committee at UHCL, 2020- 2023
- Member, DS Waiver Committee at UHCL, 2020 - 2023
- Member, CS Cybersecurity Committee at UHCL, 2017-2023
- Member, Writing Committee of Computing Sciences at UHCL, 2017-2019
- Member, CS Course Committee at UHCL, 2016-2023